Solution Design Document (SDD)

Project 4: R-Block Financial Reporting Automation | Automation Anywhere A360

Purpose: Describe the proposed technical solution for extracting, validating, storing and distributing financial reports across multiple company codes. Exact selectors, source-system steps and report-specific validation rules must be confirmed during discovery.

1. Solution Overview

The solution uses an A360 orchestration bot to load configuration, process each approved company code, retrieve the required financial reports, validate outputs, save files and distribute only eligible reports. Shared processing logic is separated from source-system-specific steps so that report navigation changes can be maintained independently.

2. Proposed Architecture

Component	Responsibility
A360 Control Room / scheduler	Schedules runs, manages bot deployment and provides centralized execution monitoring.
Main orchestration bot	Initializes the run, loads configuration, loops through reports/company codes and sets the final status.
Source-system adapter	Handles login, navigation, report parameters, execution and download/export for the approved source.
Configuration and mapping	Supplies company codes, report names, periods, output settings and validation rules.
Validation module	Checks output existence, naming, expected period and agreed completeness conditions.
Output repository	Stores validated reports and exception outputs in an access-controlled location.
Distribution module	Sends or publishes eligible reports through the approved delivery channel.
Credential vault	Provides runtime credentials without hard-coding passwords in the bot.

3. Technical Workflow

1. Scheduler starts the main bot and assigns a unique run ID.
2. Initialize timestamps, environment details, configuration and run status.
3. Validate mandatory inputs and the approved company-code/report mapping.
4. Establish a session with the approved source system using secure credentials.
5. For each report and company code, enter the configured parameters and run the report.
6. Wait for a confirmed completion state before attempting to download or export the output.

7. Save the output to a controlled temporary location and validate the file.
8. Move validated files to the final repository using the agreed naming convention.
9. Record failed, missing, duplicate or invalid outputs in an exception register.
10. Build the run summary and determine whether the overall run is successful, partially successful or failed.
11. Distribute only outputs that meet the approved validation rules.
12. Record distribution status, close application sessions and finalize logs.

4. Suggested Configuration Fields

Field	Purpose
Report_ID	Stable identifier for each configured report.
Report_Name	Business-approved report name.
Source_System	Identifier for the approved finance system or reporting source.
Company_Code	Company code to process.
Reporting_Period	Configured period or date range.
Enabled_Flag	Indicates whether the report/company-code combination is in scope for the run.
Output_Format	Approved export format, such as XLSX or CSV where supported.
Output_Path	Approved secure destination for the report.
File_Name_Pattern	Standard naming rule including report, company code and period.
Validation_Rule_Set	Identifier for report-specific validation rules.
Distribution_Group	Approved recipient group or destination.
Retry_Limit	Maximum permitted retries for recoverable failures.

5. Proposed A360 Bot Structure

Bot / task	Main responsibility
BC_Main_RBlock_Reporting	Orchestration, configuration validation, loops, overall status and cleanup.
BC_Login_Source_System	Secure login and session readiness checks.
BC_Extract_Financial_Report	Select report and company code, set period, execute and export.
BC_Validate_Output	Check file presence, size, naming, period and agreed

	report-specific conditions.
BC_Save_And_Organize_Files	Apply naming convention and move eligible outputs to the final repository.
BC_Distribute_Reports	Deliver validated reports to approved recipients and capture delivery outcomes.
BC_Log_And_Notify	Maintain run log, exception register and completion/failure notifications.

These are suggested task names. The implementation can follow the team's existing A360 naming standards.

6. Output Validation and Data Handling

- Check that the expected file exists and is not empty before marking extraction successful.
- Verify report name, company code and reporting period using available metadata or report content.
- Apply finance-approved checks such as required columns, expected sheet names or control totals only when those rules are supplied and confirmed.
- Do not alter values or recalculate financial results unless a separately approved requirement specifies the logic.
- Record validation status and reason for every rejected or incomplete output.
- Avoid overwriting an existing report without an explicit versioning or replacement rule.

7. File Naming and Storage

Proposed pattern: ReportName_CompanyCode_ReportingPeriod_RunID.ext. The exact convention should be approved by Finance. Temporary files should be kept separate from validated outputs, and final files should be stored only in an approved access-controlled repository.

- Use a consistent date and period format.
- Keep temporary downloads separate from final reports.
- Apply least-privilege permissions to folders and distribution locations.
- Follow organizational retention, backup, encryption and secure deletion policies.

8. Error Handling and Recovery

Condition	Technical handling
Source system timeout	Use bounded waits and configured retries; mark the affected item failed if unresolved.
Authentication failure	Stop protected actions, log a sanitized error and notify support.
Report still processing	Wait for a confirmed completion signal; do not assume the report is ready after a fixed delay alone.
Missing or invalid output	Mark validation failed, retain diagnostic details as permitted and exclude the file from distribution.
Company-code mapping issue	Skip the affected code and route it for finance review.

Unexpected UI change	Validate expected controls before interaction; stop the affected transaction if the layout is not recognized.
Partial run	Continue with other company codes when safe and report per-code outcomes clearly.
Distribution error	Retain the validated file in the secure repository and alert support for follow-up.
Interrupted execution	Use run IDs and processing checkpoints where feasible to support restart without accidental duplicate delivery.

9. Security and Controls

- Use approved service accounts and least-privilege access.
- Store credentials in the approved credential vault or Control Room credential facility.
- Never hard-code passwords, tokens or other secrets in bot logic or plain-text configuration.
- Restrict financial reports to authorized recipients and secure storage locations.
- Mask sensitive information in logs and avoid recording full report contents unnecessarily.
- Maintain an audit trail of run ID, company code, report, timestamps, validation result and distribution status.
- Use approved change management for bot packages, configuration and production deployment.

10. Logging and Monitoring

Event	Minimum log details
Run start	Run ID, bot version, environment and timestamp.
Report processing	Report ID, company code, reporting period and processing status.
Extraction result	Output filename, output size where available, timestamp and validation result.
Exception / retry	Step, attempt count, sanitized error and affected company code/report.
Distribution	Output identifier, recipient group or destination and delivery status.
Run completion	End time, total processed, successful, failed and exception counts.

11. Testing Strategy

Test type	Coverage
Component testing	Configuration loading, company-code mapping, file naming and validation logic.

Source-system integration	Login, report selection, period parameters, execution status and export.
Negative testing	Invalid code, unavailable system, failed export, empty file and unexpected UI changes.
End-to-end testing	Extraction through validation, storage, distribution and logging.
Security testing	Credential handling, folder permissions, recipient restrictions and log review.
Recovery testing	Retry limits, partial-run handling and restart without duplicate delivery.
UAT	Finance verifies report completeness, accuracy, company-code coverage and usability.
Performance measurement	Compare cycle time with the agreed manual baseline using the same report and company-code scope.

12. Deployment and Support

13. Confirm source systems, report list, company codes, access method and validation rules.
14. Build the bot and test components in the development environment.
15. Complete integration testing with approved test data and credentials.
16. Run UAT with Finance and obtain documented approval.
17. Promote the approved package and configuration using the production change process.
18. Verify initial scheduled runs, output access, distribution and monitoring alerts.
19. Hand over support instructions, known limitations, exception procedures and ownership.

13. Design Items Requiring Confirmation

- Exact source application(s) and whether reports are generated through UI, API or scheduled exports.
- Final company-code list, report catalogue and mapping ownership.
- Report-specific validation checks and whether totals or record counts are available.
- Output format, file naming, versioning and secure repository.
- Distribution channel, recipient groups and whether any human approval is required before sending.
- Schedule, runtime environment, retry limits, support contacts and retention rules.
- Baseline measurement and the method used to verify the target 50% cycle-time reduction.

14. Definition of Done

- Approved reports and company codes are configured and tested.
- The bot detects missing or invalid outputs and records exceptions.
- Validated reports are saved and distributed only through approved destinations.
- Security, audit logging, UAT and deployment approvals are complete.
- Cycle-time results are measured and documented against the agreed baseline.